

Minkyoo Song

✉ Mail |  LinkedIn |  Google Scholar |  CV Site
last update: September 2026

RESEARCH INTEREST

LLM Security, AI for Security, Data-driven Security, Social Network Analysis

EDUCATION

- **Korea Advanced Institute of Science and Technology (KAIST)** March 2023 - February 2027 (estimated)
Ph.D. Student in Electrical Engineering, Network and System Security Lab (Advisor: [Seungwon Shin](#)) Daejeon, South Korea
- **Korea Advanced Institute of Science and Technology (KAIST)** March 2021 - February 2023
M.S. in Electrical Engineering, Network and System Security Lab (Advisor: [Seungwon Shin](#)) Daejeon, South Korea
- **Korea Advanced Institute of Science and Technology (KAIST)** March 2016 - February 2021
B.S. in Industrial and Systems Engineering, double majored in Electrical Engineering Daejeon, South Korea

PUBLICATIONS [C]: CONFERENCE, [J]: JOURNAL, [U]: UNDER REVIEW

- [C] Y. Kim, M. Song, S. Shin. **SkillMutator: Benchmarking and Defending Language-and-Code Cross-modal Attacks on LLM Agent Skills**. 2026 Annual Computer Security Applications Conference (ACSAC 2026)
- [C] W. Choi*, M. Seo*, M. Song, H. Heo, S. Shin, M. You. **PC²: Politically Controversial Content Generation via Jailbreaking Attacks on GPT-based Text-to-Image Models**. 33rd ACM Conference on Computer and Communications Security (CCS 2026)
- [C] J. Kim, M. Song, S. Shin, S. Son. **SafeMoE: Safe Fine-Tuning for MoE LLMs by Aligning Harmful Input Routing**. The Fourteenth International Conference on Learning Representations (ICLR 2026)
- [C] J. Kim, S.H. Na, M. Song, S. Shin, S. Son. **MoEvil: Poisoning Expert to Compromise the Safety of Mixture-of-Experts LLMs**. 2025 Annual Computer Security Applications Conference (ACSAC 2025) *(Distinguished Paper Award)*
- [C] M. Song, H. Kim, J. Kim, S. Shin, S. Son. **Refusal Is Not an Option: Unlearning Safety Alignment of Large Language Models**. 34th USENIX Security Symposium (SECURITY 2025)
- [C] H. Kim, M. Song, S.H. Na, S. Shin, K. Lee. **When LLMs Go Online: The Emerging Threat of Web-Enabled LLMs**. 34th USENIX Security Symposium (SECURITY 2025)
- [C] M. Song, H. Kim, J. Kim, Y. Jin, S. Shin. **Claim-Guided Textual Backdoor Attack for Practical Applications**. The 2025 Annual Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics (NAACL 2025 Findings)
- [C] J. Kim, M. Song, S.H. Na, S. Shin. **Obliviate: Neutralizing Task-Agnostic Backdoors within the Parameter-Efficient Fine-Tuning Paradigm**. The 2025 Annual Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics (NAACL 2025 Findings)
- [C] M. Song, E. Jang, J. Kim, S. Shin. **Covering Cracks in Content Moderation: Delexicalized Distant Supervision for Illicit Drug Jargon Detection**. 31st ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD 2025)
- [C] J. Kim, M. Song, M. Seo, Y. Jin, S. Shin. **PassREfinder: Credential Stuffing Risk Prediction by Representing Password Reuse between Websites on a Graph**. 2024 IEEE Symposium on Security and Privacy (SP) (Oakland 2024)
- [J] J. Kim, M. Song, M. Seo, Y. Jin, S. Shin, J. Kim. **PassREfinder-FL: Privacy-Preserving Credential Stuffing Risk Prediction via Graph-Based Federated Learning for Representing Password Reuse between Websites**. Elsevier Expert Systems with Applications (ESWA)
- [J] J. Choi, J. Kim, M. Song, H. Kim, N. Park, M. Seo, Y. Jin, S. Shin. **A Large-Scale Bitcoin Abuse Measurement and Clustering Analysis Utilizing Public Reports**. IEICE Transactions on Information and Systems
- [U] M. Song, J. Kim, M. Kang, H. Kim, S. Shin, S. Son. **Graphs Don't Stay Secret: Practical Subgraph Reconstruction Attacks on Defended Graph RAG**. Under Review
- [U] K. Kim, J. Cui, M. Song, S. Shin. **Exploring the Familiar Taste of Toxicity: A Causal Influence Analysis of Toxic Comments on Internet Forums**. Invited to Major Revision at IEEE Transactions on Knowledge and Data Engineering (TKDE)

- [U] H. Kim, J.Cui, **M. Song**, H. Heo, S. Shin, K. Lee, X. Liao. **SCRIPTIOC-BENCH: A Benchmark for Recognizing Actionable Threat Intelligence from Script-Based Malware using LLMs.** *Under Review*
- [U] K. Kim, S.H. Na, **M. Song**, S. Shin. **Global Meta-path-level Counterfactual Explanation for Heterogeneous Graph Neural Networks by Path Exclusion.** *Under Review*
- [U] J. Kim, M. Seo, **M. Song**, S. Shin, J. Kim. **To Make Each Account Count: Exploring Credential Data Breach Threats through Victim-driven Analysis.** *Under Review*

EXPERIENCE

- **Johns Hopkins University** September 2026 - February 2027
Visiting Scholar
 ◦ Working with [Prof. Yinzhi Cao](#). Baltimore, MD, USA
- **S2W**  July 2022 - Feb 2023
Research Intern @ AI Team South Korea
 ◦ **Illicit drug jargon detection:** Analyzed illicit drug-related discussions and developed an LLM-based content moderation framework, independently capturing contextual and lexical characteristics.

HONORS AND AWARDS

- **Distinguished Paper Award** 2025
Annual Computer Security Applications Conference (ACSAC)
 ◦ MoEvil: Poisoning Expert to Compromise the Safety of Mixture-of-Experts LLMs
- **4th Prize, 2025 Cybersecurity Paper Competition** 2025
Korean Association of Cybersecurity Studies (KACS)
 ◦ Poisoning Expert to Compromise the Safety of Mixture-of-Experts LLMs
- **2nd Prize, 2023 Cybersecurity Paper Competition** 2023
Korean Association of Cybersecurity Studies (KACS)
 ◦ Graph-based Deep Learning Framework for Credential Stuffing Risk Prediction
- **4th Prize, 2023 Cybersecurity Paper Competition** 2023
Korean Association of Cybersecurity Studies (KACS)
 ◦ Delexicalized Distant Supervision for Illicit Drug Jargon Detection
- **4th Prize, 2023 Cybersecurity Paper Competition** 2023
Korean Association of Cybersecurity Studies (KACS)
 ◦ Understanding the Occurrence and Impact of Credential Data Breach
- **Cum Laude** 2021
Korea Advanced Institute of Science and Technology (KAIST)
- **Academic Achievement Award: Salutatorian** 2019 Spring
Korea Advanced Institute of Science and Technology (KAIST)
- **Dean's List** 2019 Spring
Industrial and Systems Engineering (ISysE, KAIST)

LANGUAGES

Korean (Native), English (Fluent)